Probability Rules

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Rules of probability

- Probability addition rule
- Difference Rule
- Inclusion Rule/Exclusion Rule
- Complement Rule

Addition Rule

- This rule states that the number of elements in a union of mutually disjoint finite sets equals the sum of the number of elements in each of the component sets.

$$N(A) = N(A1)+N(A2)+\ldots+ N(Ak)$$

- Example

A computer access password consists of from one to three letters chosen from the 26 in the alphabet with repetitions allowed. How many different passwords are possible?

Solution:

- By the addition rule, the total number of passwords equals the number of passwords of length 1, plus the number of passwords of length 2, plus the number of passwords of length 3.

Number of passwords of length 1= 26

Number of passwords of length 2 $=26^2$

Number of passwords of length 2 $=26^3$

Total no of passwords = 26 + 26^2 $+26^3$

= 18278

Difference Rule

- if the number of elements in a set A and the number in a subset B of A are both known,

Number of elements that are in A and not in B can be computed.

The Difference Rule:

If A is finite set and B is a subset of A, then

$$N(A-B) = N(A) - N(B)$$

Example

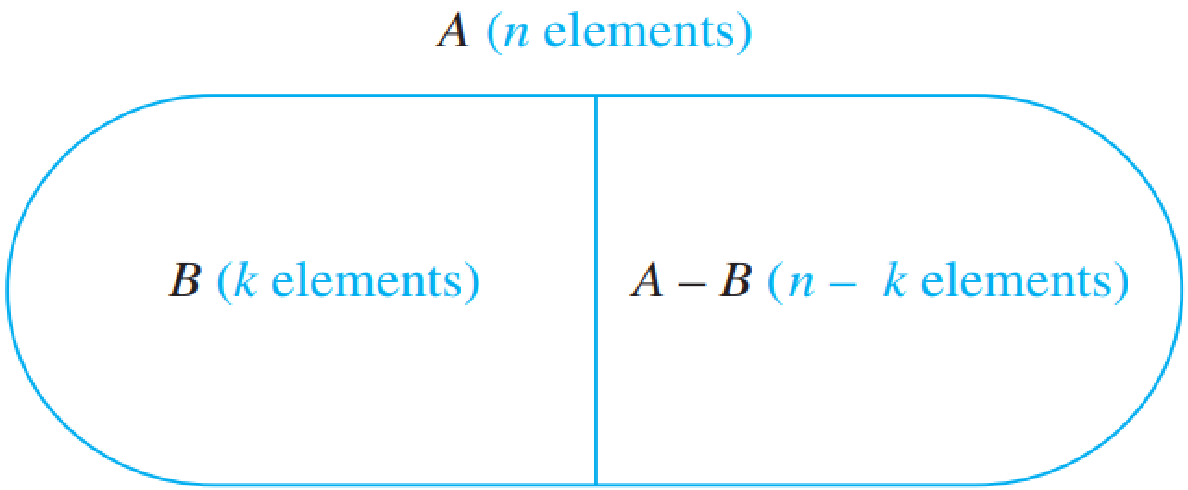


Figure 9.3.3 The Difference Rule

The difference rule holds for the following reason: If B is a subset of A, then the two sets B and A − B have no elements in common and B ∪ (A − B) = A.

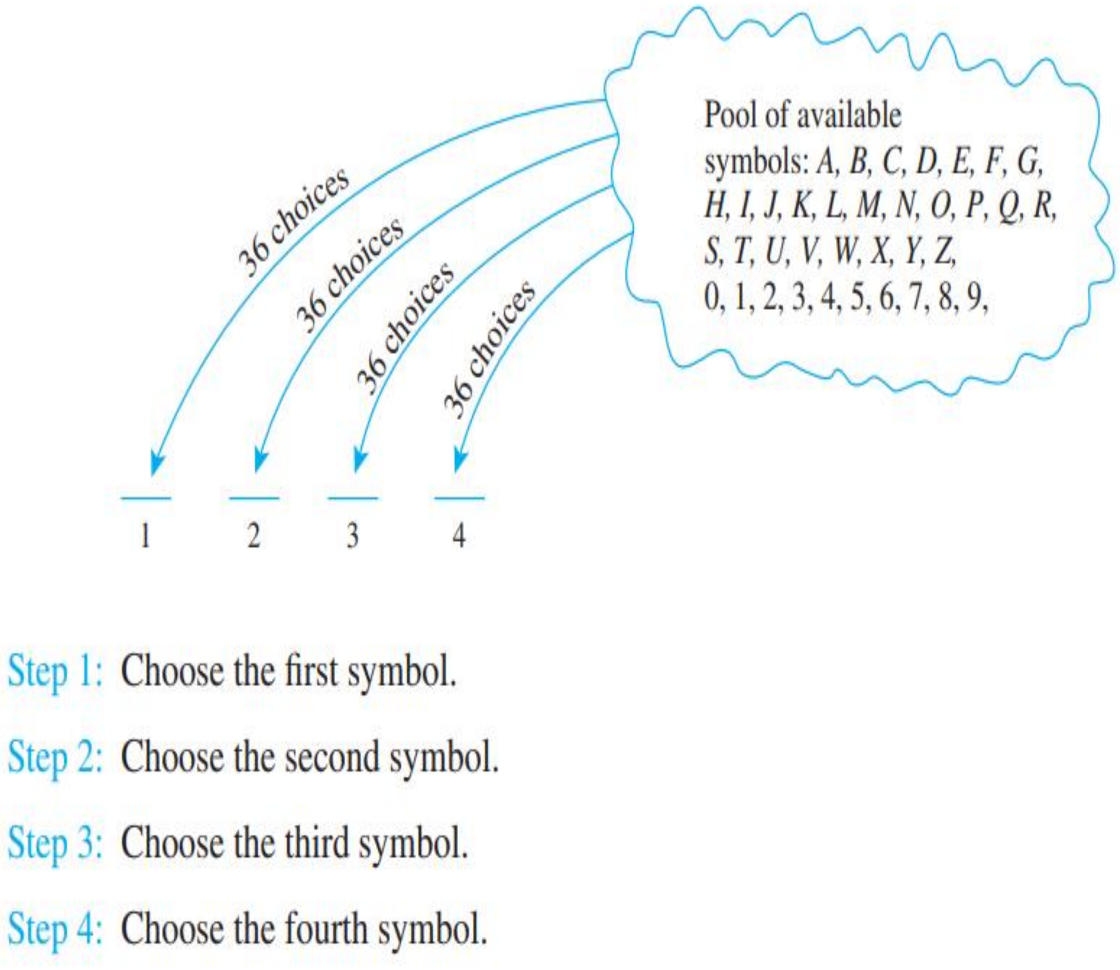
Hence, by the addition rule, N(B) + N(A − B) = N(A).

Example

- Example:

A typical PIN (personal identification number) is a sequence of any four symbols chosen from the 26 letters in the alphabet and the ten digits, with repetition allowed.

1. How many PINs contain repeated symbols?
2. If all PINs are equally likely, what is the probability that a randomly chosen PIN contains a repeated symbol?



Part A

- The total number of possible PINs is the number of choices for each symbol raised to the power of the number of symbols in a PIN.
- In this case, there are 36 choices (26 letters + 10 digits) for each symbol, and a PIN consists of 4 symbols.

$$36^4 = 1{,}679{,}616.$$

- Pins When No Repition allowed

$$= 36 * 35*34* 33$$

$$= 265896$$

If all PINs are equally likely, what is the probability that a randomly chosen PIN contains a repeated symbol?

P (E) =

E = Pins Contain a Repeated Symbol

S = total sample space

Event of interest = E/S

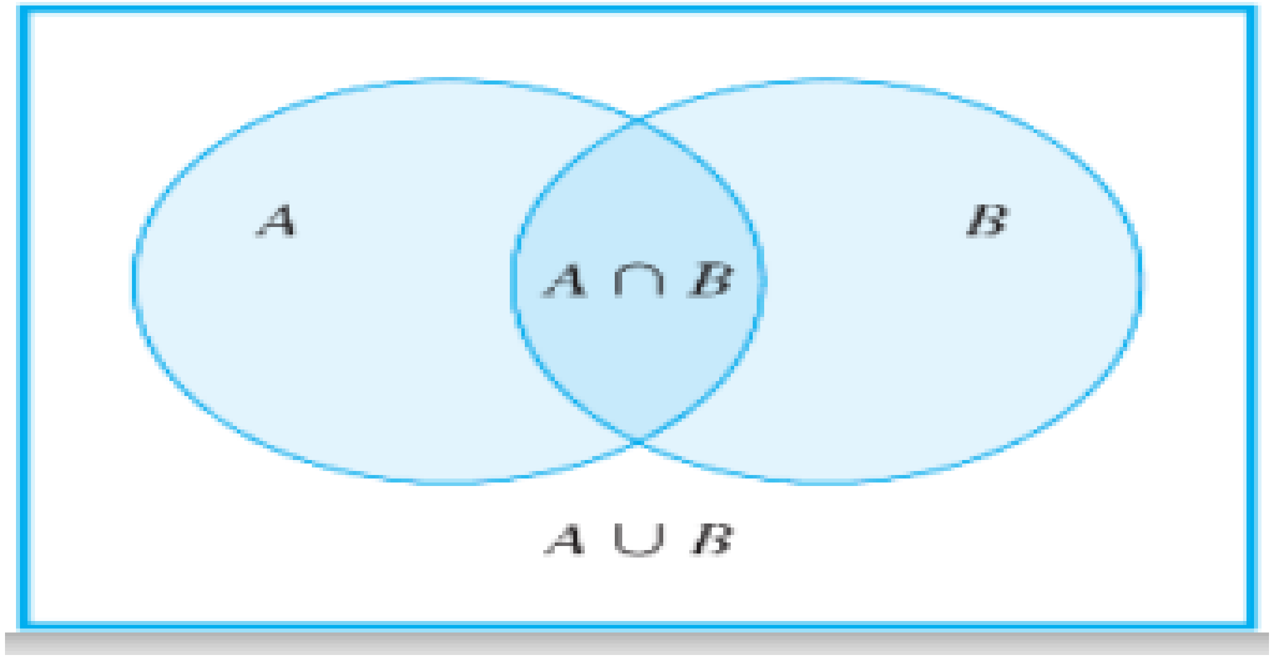
=265896/1,679,616.

=0.158

=15.8%

Inclusion and exclusion Rule

- The addition rule says how many elements are in a union of sets if the sets are mutually disjoint. Now consider the question of how to determine the number of elements in a union of sets when some of the sets overlap.

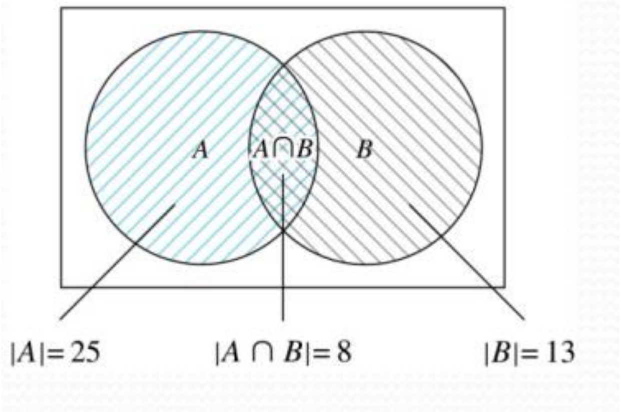


Example:

- In a discrete mathematics class every student is a major in computer science or mathematics or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in the class?

Solution

$|A \cup B| = |A| + |B| - |A \cap B| = 25 + 13 - 8 = 30$



Example

- How many integers from 1 through 1,000 are multiples of 3 or multiples of 5?
- How many integers from 1 through 1,000 are neither multiples of 3 nor multiples of 5?

Solution

- Let A = the set of all integers from 1 through 1,000 that are multiples of 3.
- Let B = the set of all integers from 1 through 1,000 that are multiples of 5.
- $A \cup B$ = the set of all integers from 1 through 1,000 that are multiples of 3 or multiples of 5
- $A \cap B$ = the set of all integers from 1 through 1,000 that are multiples of both 3 and 5 = the set of all integers from 1 through 1,000 that are multiples of 15

General Calculations

Multiples of 3 = 3k
3 *333 = 999
Its means 333 elements having multiple of 3.
Multiple of 5 = 5k
5 * 200.

N(A) = 333
N(B) = 200

Part A

3k and 5k =15k
15 * 66 = 990
$N(A \cup B) = N(A) + N(B) - N(A \cap B)$
$= 333 + 200 - 66$
$= 467$

Part B

3k and 5k =15k
15 * 66 = 990
$1000 - N(A \cup B)$
1000 – 467 = 533.

Solution

$$N(A \cup B) = N(A) + N(B) - N(A \cap B)$$

$$= 333 + 200 - 66$$

$$= 467.$$

Counting Rules in Term of Probability

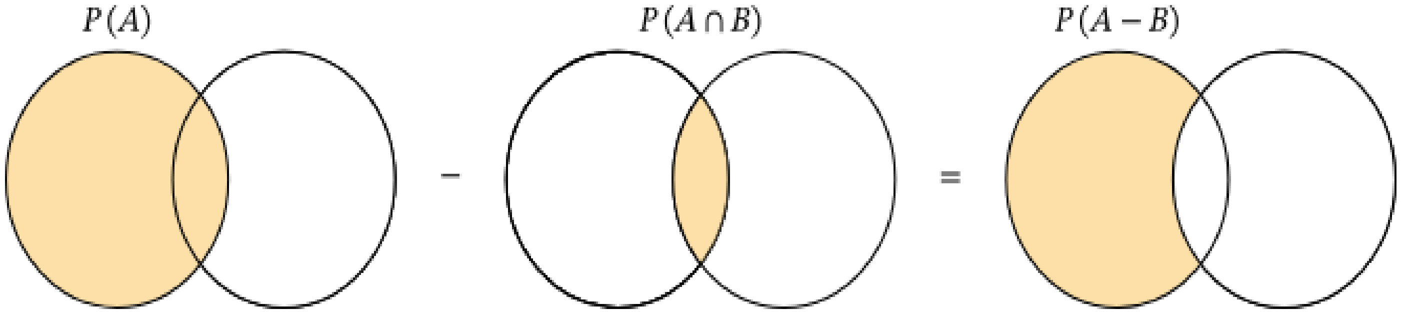
Rules

- Counting Difference Rule
- Counting Inclusion exclusion

Probability Difference Rule

- The probability of the difference between two events A and B

$$P(A-B)=P(A)-P(A\cap B).$$



Example

- A ball is drawn at random from a bag containing 12 balls each with a unique number from 1 to 12. Suppose A is the event of drawing an odd number and B is the event of drawing a prime number. Find $P(A-B)$.

Probability Inclusion/Exclusion Rule

$$\Pr \boldsymbol{A} \cup \boldsymbol{B} = \boldsymbol{Pr}\ \boldsymbol{A} + \boldsymbol{Pr}[\boldsymbol{B}] - \Pr[\boldsymbol{A} \cap \boldsymbol{B}]$$

- Example:

Suppose A and B are two events with probabilities

$$P(A)=57 \text{ and } P(B)=47.$$

Given that $P(A \cup B)=67$,determine $P(A-B)$.

Probability

- A countable samples space S is a nonempty countable set. An element $w \in S$ is called an outcome. A subset of S is called an event.
- A probability function on a sample space S is total function $Pr: S \to \mathscr{R}$ such that

$$\Pr w \geq 0, for\ all\ w \in S \text{ and}$$

$$\sum_{w \in S} \Pr w = 1$$

- A sample space together with a probability function is called probability space. For any event $E \subseteq S$, the **probability** of E is defined to be the sum of the probabilities of the outcomes in E.

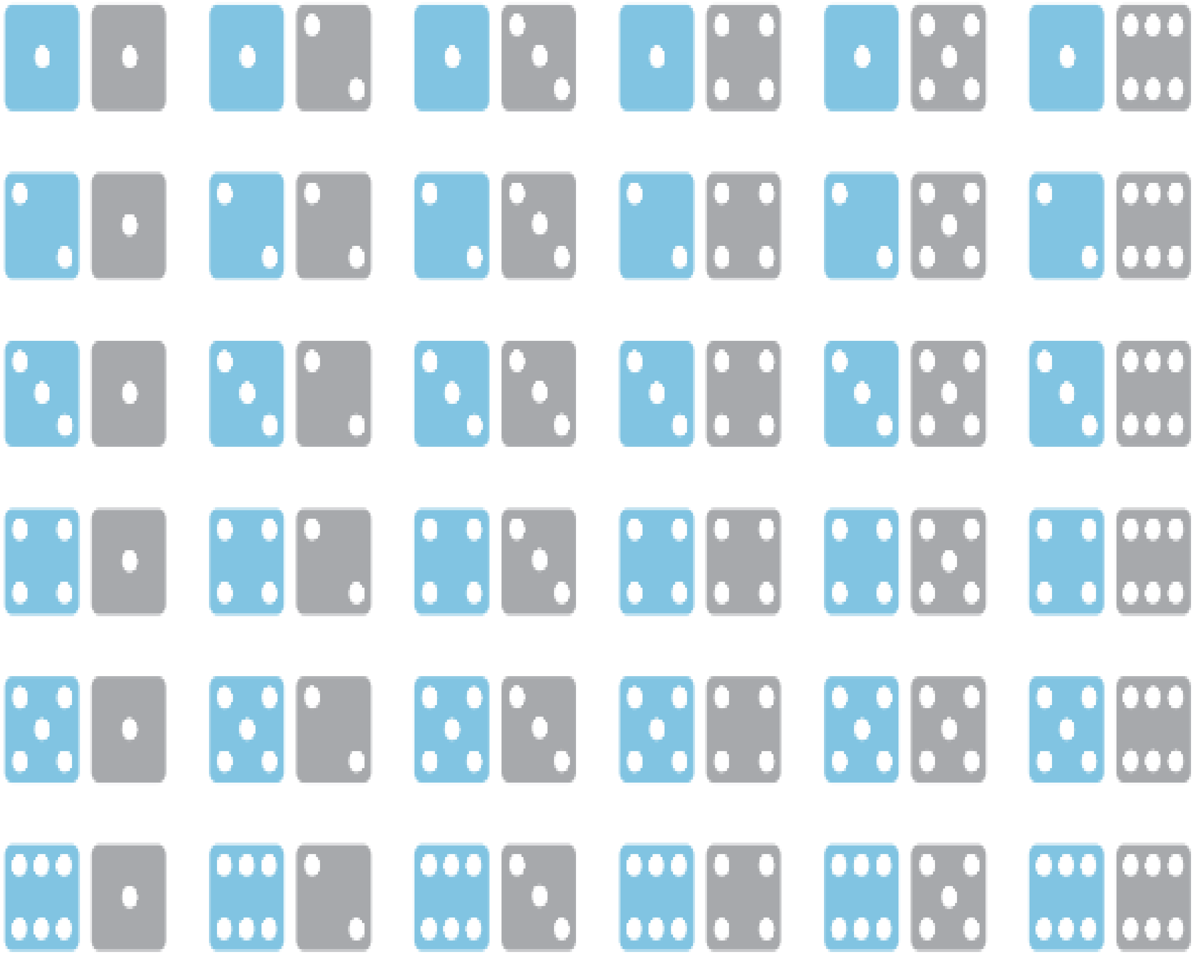
$$\Pr E ::= \sum_{w \in S} pr[w]$$

Probability function

- A probability function P from the set of all events in S to the set of real numbers satisfies the following three axioms: For all events A and B in S,

1. $0 \leq P(A) \leq 1$
2. $P(\emptyset) = 0$ and $P(S) = 1$
3. If A and B are disjoint (that is, if $A \cap B = \emptyset$), then the probability of the union of A and B is

$$P(A \cup B) = P(A) + P(B).$$



Use the compact notation to write the sample space S of possible outcomes.

Use set notation to write the event E that the numbers showing face up have a sum of 6 and find the probability of this event.

Solution

- S = {11, 12, 13, 14, 15, 16, 21, 22, 23, 24, 25, 26, 31, 32, 33, 34, 35, 36, 41, 42, 43, 44, 45, 46, 51, 52, 53, 54, 55, 56, 61, 62, 63, 64, 65, 66}.
- E = {15, 24, 33, 42, 51}.

$$\mathbf{Pr}\ [\boldsymbol{E}] = |\boldsymbol{E}|\ /\ |\boldsymbol{S}|$$

Permutation

- A permutation is the choice of r things from a set of n things without replacement and where the order matters.
- Example

A permutation of a set S is a sequence that contains every element of S exactly once.

Let S = {a, b, c},

then here are all its permutations

$$(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c})\ (\boldsymbol{a}, \boldsymbol{c}, \boldsymbol{b})\ (\boldsymbol{b}, \boldsymbol{a}, \boldsymbol{c})\ (\boldsymbol{b}, \boldsymbol{c}, \boldsymbol{a})\ (\boldsymbol{c}, \boldsymbol{a}, \boldsymbol{b})\ (\boldsymbol{c}, \boldsymbol{b}, \boldsymbol{a})$$

Combination

- In Combination order of arrangement is not important.
- Combinations are used when the same kind of things are to be sorted.

$$_nC_r = \frac{n!}{r!(n-r)!}$$

Example

- we select 5 cards at random from a deck of 52 cards. What is the probability that we will end up having a full house?

Solution

- A full house consists of three cards of one rank and two cards of another rank. There are 13 ranks in a deck of cards (Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King), and for each rank, there are 4 different suits (hearts, diamonds, clubs, spades).

Solution (cont)

- Number of ways to choose three cards of one rank:
- There are 13 ranks to choose from, and we need to select 3 of them. So, the number of ways to choose three cards of one rank is given by the combination formula:
- C(13, 1) * C(4, 3)
- = 13 * 4 = 52.

- After selecting three cards of one rank, we have 12 remaining ranks to choose from. We need to select 2 of them. So, the number of ways to choose two cards of another rank is given by the combination formula:
- C(12, 1) * C(4, 2)
- = 12 * 6
- = 72

Other Types Of Probaility

- Conditional Probability
- Independence Probability
- Total Probability
- Bayes Theoram

Conditional Probability

- Let A and B be events in a sample space S. If $P(A) = 0$, then the conditional probability of B given A, denoted $P(B \mid A)$, is

$$P(A \mid B) = P(A \cap B) / P(B) \ .$$

Halting Problem

- Imagine a best-of-three tournament and the following scenario

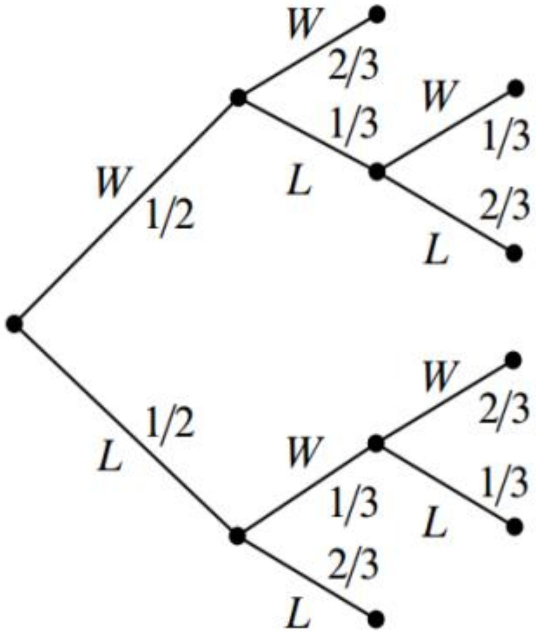
1. The Halting problem wins the first game with probability 1/2
2. If they won the previous game, they win the next game with probability 2/3
3. If they lose the previous game, the next game is won with probability 1/3
4. What is the probability that halting problem will win the tournament?
5. Is this a question about conditional probability?

Solution

- Let The halting problem wins the tournament
  - The halting problem won the first game
  - What we want to know is the conditional probability
  - $\boldsymbol{P}\ [\boldsymbol{A} \mid \boldsymbol{B}]$.

The Halting Problem Tree

game 1 game 2 game 3



outcome	event A: win the series	event B: win game 1	outcome probability
WW	✓	✓	1/3
WLW	✓	✓	1/18
WLL		✓	1/9
LWW	✓		1/9
LWL			1/18
LL			1/3

Solution

- Let S be the sample space, then

$$\boldsymbol{S} = \{\boldsymbol{WWW}, \boldsymbol{WLW}, \boldsymbol{WLL}, \boldsymbol{LWW}, \boldsymbol{LWL}, \boldsymbol{LLL}\}$$

- Let T be the event that the halting problem wins the tournament, then

$$\boldsymbol{T} = \{\boldsymbol{WWW}, \boldsymbol{WLW}, \boldsymbol{LWW}\}$$

- And F be the event that they win the first game, then

$$\boldsymbol{F} = \{\boldsymbol{WWW}, \boldsymbol{WLW}, \boldsymbol{WLL}\}$$

Solution (Cont)

- Then $\boldsymbol{P}\ [\boldsymbol{A}\ |\ \boldsymbol{B}]$ (probability that the halting problem wins the tournament given that they win their first game), can be computed as

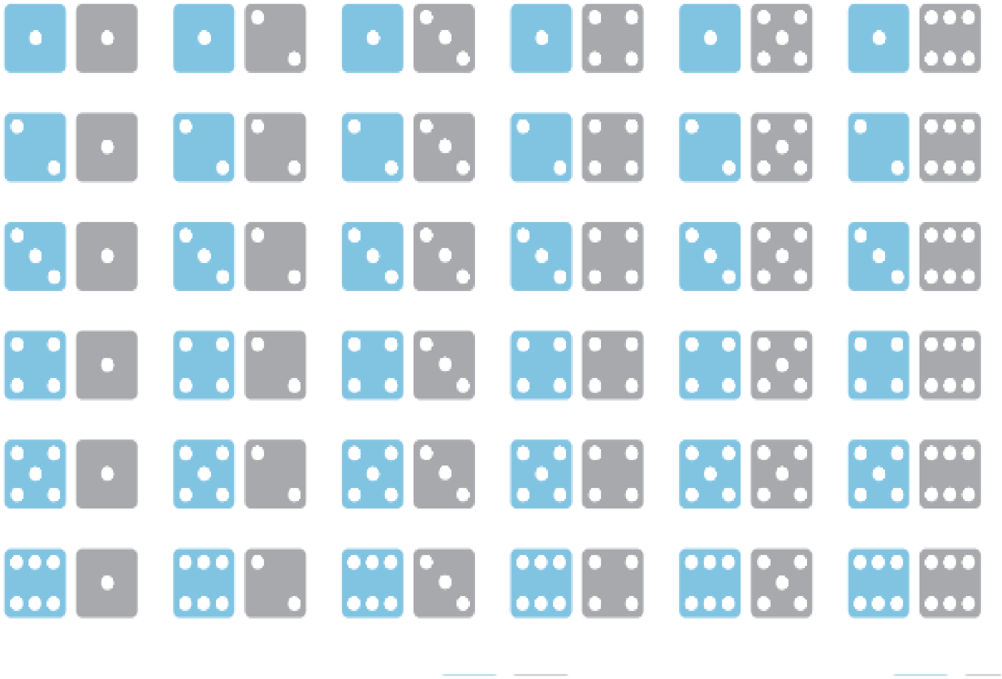
$$\boldsymbol{Pr}\ [\boldsymbol{A}\ |\boldsymbol{B}] ::= \mathbf{Pr}\ [\boldsymbol{A} \cap \boldsymbol{B}] /\ \mathbf{Pr}\ [\boldsymbol{B}]$$

$$= \boldsymbol{Pr}\ [\boldsymbol{WWW}, \boldsymbol{WLW}]\ /\boldsymbol{Pr}\ [\boldsymbol{WWW}, \boldsymbol{WLW}, \boldsymbol{WLL}]$$

$$= \frac{\frac{1}{3} + \frac{1}{18}}{\frac{1}{3} + \frac{1}{18} + \frac{1}{9}} = \frac{7}{9}$$

Example

A pair of fair dice, one blue and the other gray, are rolled. What is the probability that the sum of the numbers showing face up is 8, given that both of the numbers are even?



Solution

- The sample space is the set of all 36 outcomes obtained from rolling the two dice and noting the numbers showing face up on each.
- Let A be the event that both numbers are even

$$A = \{22, 24, 26, 42, 44, 46, 62, 64, 66\},$$

- B the event that the sum of the numbers is 8.

$$B = \{26, 35, 44, 53, 62\}$$

- $A \cap B$

$$\{26, 44, 62\}.$$

$$P(A) = 9/36$$

$$P(B) = 5/36$$

$$P(A \cap B) = 3/36.$$

- By definition of conditional probability,

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{3}{36}}{\frac{9}{36}} = \frac{3}{9} = \frac{1}{3}.$$

Product Rule

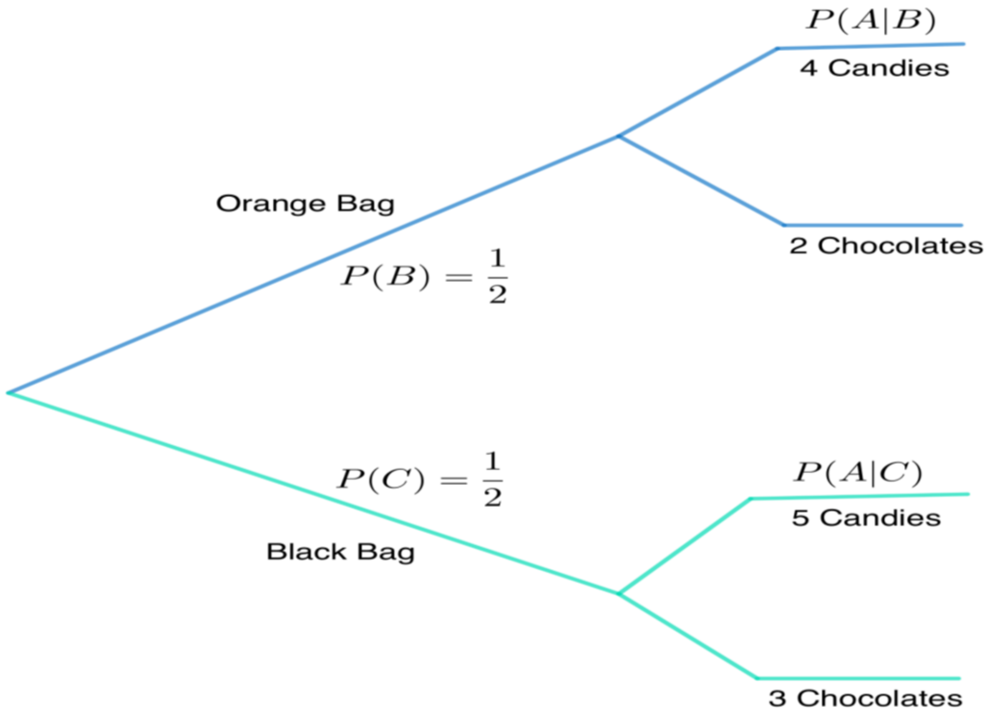
$$P(A \cap B) = P(A \mid B) * P(B)$$

Example:

There are 2 bags, an orange bag and a black bag. There are 4 candies in the orange bag and 5 candies in the black bag. There are also 2 chocolates in the orange bag and 3 chocolates in the black bag. A sweet is randomly chosen, find the probability that the chosen sweet is a candy and it came from the black bag.

Solution:

- Let A be the event that the chosen sweet is a candy and let B be the event that the sweet was chosen from the orange bag. Let C be the event that the bag chosen was the black one.



We want to find the probability that the sweet chosen is a candy given that the bag is black, hence we want to find $P(A \cap C)$.

Using the product rule we have,

$$P(A \cap C) = P(A|C)P(C)$$

The probability that the sweet came from the black bag is

$$P(A \cap C) = \frac{candiesintheblackbag}{totalsweetsinblackbag} = \frac{5}{8}$$

and the probability of choosing the black bag is 1/2 as there are only two bags,

$$P(C) = \frac{1}{2}$$

Substituting these values, we get,

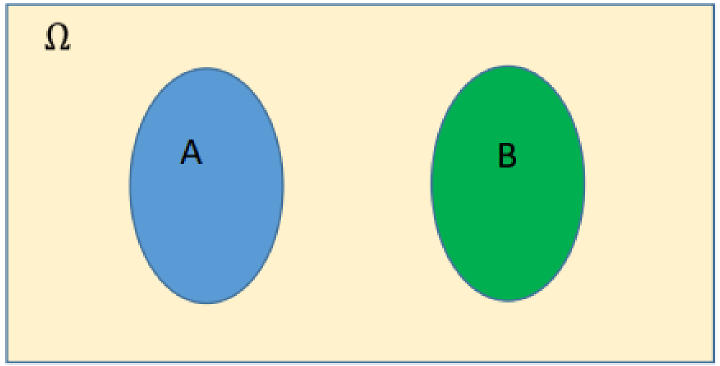
$$P(A \cap C) = \frac{5}{8} \cdot \frac{1}{2} = \frac{5}{16}$$

Hence, the probability that the sweet is a candy and it came from the black bag is $\frac{5}{16}$.

Independence

- In probability, two events are independent if the incidence of one event does not affect the probability of the other event. If the incidence of one event does affect the probability of the other event, then the events are dependent.

$$P(B|A) = P(B)$$
$$P(A|B) = P(A)$$



- We know this already

$$P(A)\,P(B) = P(A \cap B)$$

.

What the Independence really means

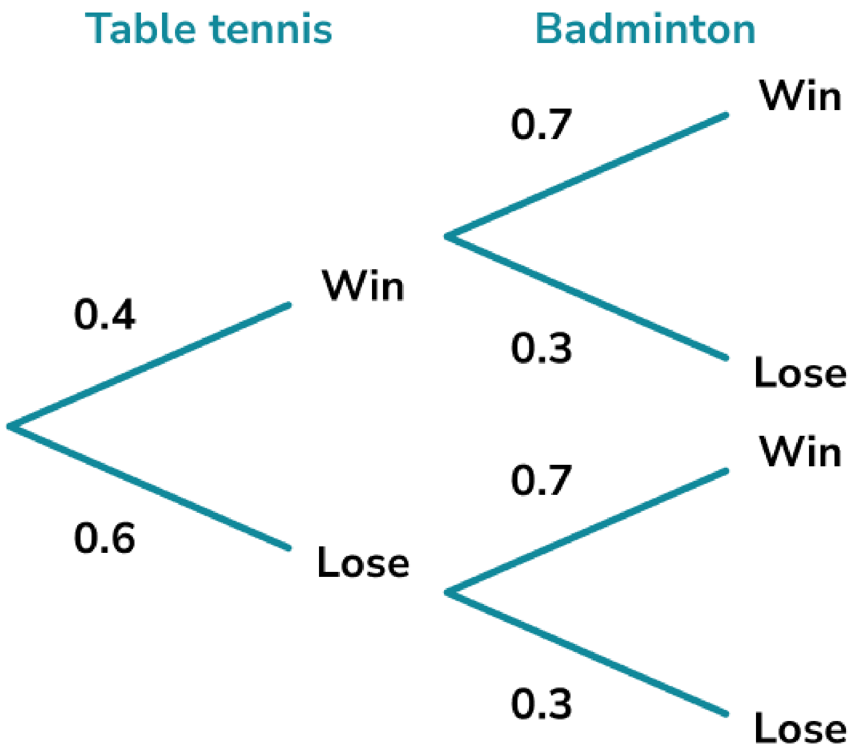
- Thus being dependent is completely different from being disjoint!
- Two events are independent, if the occurrence of one does not change our belief about the occurrence of the other
- Typically the case when the two events are determined by two physically distinct and non-interacting processes.
- Getting heads in a coin toss and snowing outside

Three Basic Steps

1. Confirm that events are independent
2. Identify the probability of event
3. Multiply the Probability

Example:

- A student plays a game of table tennis on Saturday and a game of badminton on Sunday. The different outcomes are represented on the tree diagram below.



Solution

- Step 1:

The outcome of one game does not affect the outcome of the other game therefore the events are independent.

- Step 2:

The probability that student wins table tennis is 0.4 and the probability that student wins badminton is 0.7.

- Step 3:

The probability that Student wins table tennis and badminton is

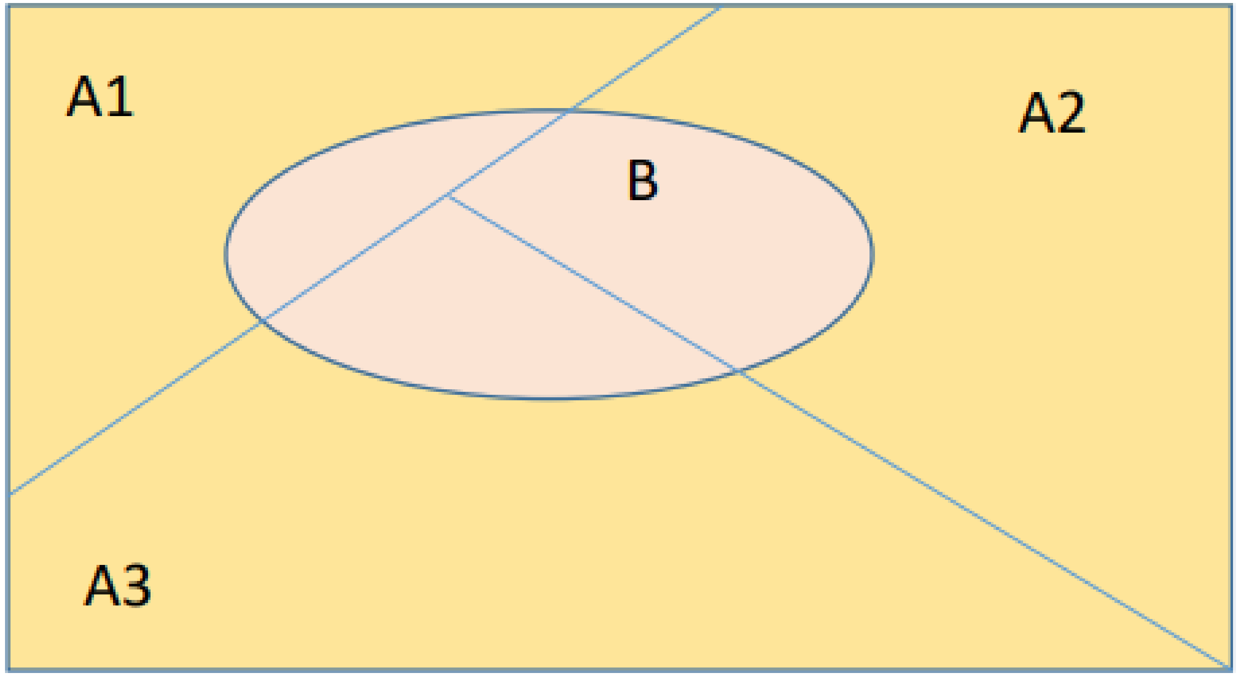
$$0.4 \times 0.7 = 0.28$$

Difference Between Dependent and Independent Event

Independent Events	Dependent Events
The first marble is replaced. X: blue first Y: yellow second P(X) × P(X/Y) = P(X and Y) $\frac{6}{15} \times \frac{9}{15} = \frac{54}{225} \approx 24\%$	The first marble is not replaced. X: blue first Y: yellow second P(X) × P(X/Y) = P(X and Y) $\frac{6}{15} \times \frac{9}{14} = \frac{54}{210} \approx 26\%$

Total Probability

Total Probability



What is the probability of P(B)?

Derive Formula

$$P(B) = (P \cap A_1) + (P \cap A_2) + (P \cap A_3)$$

$$\mathrm{P(B)} = \mathrm{P(B}|\, A_1) * (A_1) + \mathrm{P(B}|\, A_2) * (A_2) + \mathrm{P(B}|\, A_3) * (A_3)$$

$$= \sum_{I} \mathrm{P(B}|\, A_I) * (A_I)$$

Bayes Theorem

Bayes' Theorem states that the ***conditional probability of an event***, based on the ***occurrence of another event***, is equal to the likelihood of the ***second event given the first event multiplied by the probability of the first event***.

$$P(B_k \mid A) = \frac{P(A \mid B_k)P(B_k)}{P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + \cdots + P(A \mid B_n)P(B_n)}$$

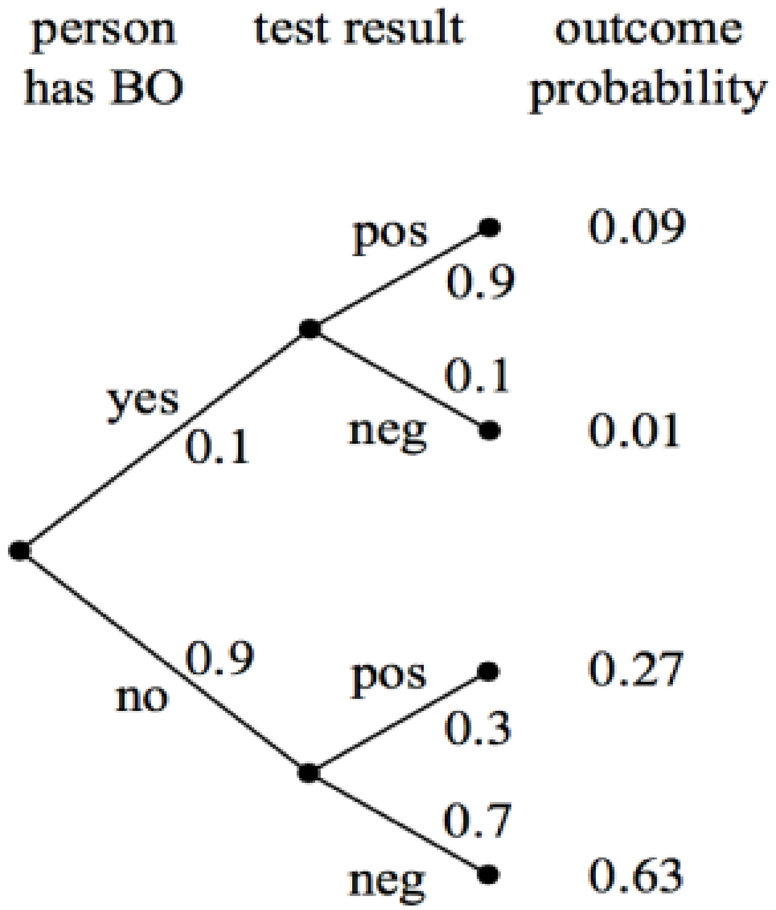
Example

- Let’s assume a “not-so-perfect” test for a medical condition called BO suffered by 10% of the population • The test is not-so-perfect because
  - 90% of the tests come positive if you have BO
  - 70% of the tests come negative if you don’t have BO

If we randomly test a person for BO, and if the test comes positive, what is the probability that the person has BO.

Solution

- A: The test came positive
- B: The person has BO



Solution (Cont…)

$$P(B|A) = \frac{(A \cap B)}{P(A)}$$

$$P(A) = P(B|\, A_p) * (A_p) + P(B|\, A_p) * (A_p)$$

$$P(A) = 0.27 + 0.09$$

$$P(B|A) = \frac{0.09}{0.09+0.27}$$

$$= \frac{1}{4}$$